At Equal Review Cost, Context Helps: A Leakage-Corrected Comparison of a Transformer and a Lexical Baseline for Propaganda Detection

Working Paper II in Computational Political Anthropology — Version 2 (corrected)

Șerban Gabriel Florin

Independent Researcher, Hannover, Germany

ORCID: 0009-0000-2266-3356

Code & data: github.com/politomorphism/politomorphism-defense

Revised November 2026 — supersedes Version 1 ("Context Buys Precision, Not Recall")

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Correction note (what changed in Version 2)

This version corrects Version 1 ("Context Buys Precision, Not Recall") after independent reproduction audits. Four changes. (1) The official PropXplain English split has article-level leakage; both models are re-run on the same article-disjoint split used in Working Paper I. (2) Version 1 compared the two models only at their default thresholds and concluded that context improved precision but "not recall." With per-paragraph scores now saved, a fair comparison reverses that conclusion: the models flag different numbers of paragraphs at their defaults, and at equal review cost the transformer's recall is in fact higher. (3) Version 1 asserted, without paired evidence, that the transformer "reclassified some fifty" particular cases and that "the same subtle cases remain missed"; these are replaced with the measured paired transitions, which show the two models miss substantially different cases. (4) The transformer was retrained across five random seeds, so its advantage is no longer a single run: it beats the baseline on average precision, on recall at matched review budget, and on paired correctness in all five seeds.

Abstract

Working Paper I established a lexical baseline for propaganda detection on an article-disjoint re-split of the English PropXplain corpus. This paper compares that baseline with a fine-tuned DistilBERT on the same split, with per-paragraph scores saved so the comparison is not confined to one operating point. At the two models' default thresholds the transformer is more precise (0.578 vs 0.464) and flags fewer paragraphs (249 vs 345), while its recall is lower (0.545 vs 0.606); the difference in F1 at these thresholds is not statistically distinguishable from zero (ΔF1 +0.033, 95% article-clustered bootstrap CI [−0.03, +0.09]). On fair, threshold-independent comparisons, however, the transformer is the better detector: its average precision is higher (ΔAP +0.074 [+0.00, +0.14]); at an equal review budget its recall exceeds the baseline's (0.670 vs 0.606; Δ +0.068 [+0.01, +0.12]); and it classifies significantly more test paragraphs correctly (McNemar, 148 vs 84 discordant, $p < 0.001$). Paired transitions show the two models miss different cases, not the same ones. The earlier conclusion that "context buys precision, not recall" does not hold on the clean split once the comparison is made at equal cost. Retraining the transformer across five random seeds confirms the advantage is not a fluke of one run: average precision is 0.605 ± 0.005, and the transformer beats the baseline on average precision, on matched-budget recall, and on paired correctness in all five seeds. The detector is still not a validated measure of a construct, and claims are scoped to classification only.

Keywords: propaganda detection; DistilBERT; matched review cost; average precision; McNemar; article-disjoint evaluation; computational political anthropology; Politomorphism

1. Introduction

Working Paper I established a transparent lexical baseline (TF-IDF with logistic regression) for propaganda detection, evaluated on an article-disjoint re-split of the English PropXplain corpus after an article-level leakage problem in the official split was corrected. This paper asks whether a context-aware model does better, and — unlike the first version of this paper — it answers that question with per-paragraph scores saved, so the two detectors can be compared not only at one arbitrary decision threshold but by their ranking and at an equal review cost. The earlier version compared only default thresholds and concluded that context "buys precision, not recall." That conclusion does not survive a fair comparison on the clean split, as shown below.

A word on what "context helps" means here, to prevent the title from being read as more than it is. This paper measures one thing: how well each model reproduces the PropXplain annotators' binary labels on a clean split of one English news corpus. The finding is that a contextual model reproduces those labels better than a lexical one when compared fairly. It is not a claim that the model detects manipulation as a validated construct, that it would transfer to other corpora, languages, or political contexts, or that "context helps" in the broad political or anthropological sense the Politomorphism framework is ultimately concerned with. This is a study of corpus-label transfer, and every result below should be read within that scope.

2. Data and method

Both models use the identical article-disjoint split of Working Paper I (train 4,366 paragraphs, validation 701, test 862 with 264 propagandistic; no article spans two sets; exact-text duplicates removed). The transformer is DistilBERT (distilbert-base-uncased) fine-tuned for three epochs with a class-weighted cross-entropy mirroring the baseline's balanced weights, learning rate 2e-5, maximum length 256 tokens. To estimate training-seed variability — which a single run cannot — the transformer is retrained with five random seeds (42, 1, 2, 3, 4) on this fixed split; the baseline is deterministic on a fixed split, so it contributes one reference point. Per-paragraph positive-class scores are saved for every run. Uncertainty on differences uses 2,000 bootstrap resamples drawn at the article level; paired correctness is compared with an exact McNemar test; recall at matched review budget flags, for the transformer, the same number of paragraphs the baseline flags, taken by score rank. Matched review budget is a device for equalizing cost between the two models — it answers "at the same number of alarms, which catches more?" — and is not a claim that a deployed system would be set to flag exactly that many paragraphs; the operating point of a real system would be chosen to meet a precision or workload target. The default-threshold tables below report the primary seed (42); the seed sweep is reported alongside.

3. Results

Table 1 reports both models at their default thresholds. The transformer is more precise and flags fewer paragraphs; its recall is lower. Table 2 reports the differences with article-clustered bootstrap intervals.

Model (propagandistic class)	Precision	Recall	F1	AP	Flagged
TF-IDF + LR (baseline)	0.464	0.606	0.525	0.527	345
DistilBERT (transformer)	0.578	0.545	0.561	0.606	249

Table 1. Default-threshold performance on the article-disjoint test set (n = 862; 264 propagandistic).

Difference (transformer − baseline)	Estimate	95% CI	Distinguishable from 0?
F1 at default thresholds	+0.033	[−0.03, +0.09]	no
Average precision (ranking)	+0.074	[+0.00, +0.14]	yes
Recall at matched review budget	+0.068	[+0.01, +0.12]	yes
Correctness (McNemar)	148 vs 84	p < 0.001	yes

Table 2. Transformer minus baseline, with 95% article-clustered bootstrap intervals (2,000 resamples). "Distinguishable from 0" means the interval excludes zero.

The picture is consistent and, read carefully, favourable to the transformer. At the default thresholds the F1 difference (0.525 → 0.561) is not statistically distinguishable from zero, and the two models sit at different operating points — the transformer flags 249 paragraphs, the baseline 345 — so comparing their default recalls is comparing different review costs. Once that is controlled, the transformer is the better detector: its average precision is higher by a margin whose interval just excludes zero (ΔAP +0.074 [+0.00, +0.14]), meaning it ranks propaganda above non-propaganda more reliably regardless of threshold; and when it is held to the same review budget as the baseline (345 flags), its recall rises to 0.670 against the baseline's 0.606 (Δ +0.068 [+0.01, +0.12]). A McNemar test on paired correctness is decisive: the transformer is right where the baseline is wrong on 148 paragraphs, against 84 the other way (p < 0.001).

3.1 Robustness across training seeds

Because a transformer's result varies with its random seed, the comparison above was repeated for five seeds on the fixed split. Table 3 summarizes the outcome. The transformer's average precision is 0.605 ± 0.005 — a very small spread — against the baseline's 0.527, and in every one of the five seeds the transformer exceeds the baseline on all three fair measures: average precision, recall at matched review budget, and paired correctness (McNemar p ≤ 0.0001 each time). Recall at the default threshold is the most seed-sensitive quantity (ranging from 0.49 to 0.62), which is why the threshold-independent and matched-cost measures, not the default recall, carry the comparison.

Seed	AP	Recall @ matched budget	McNemar (tf/base correct)	p
42	0.606	0.670	148 / 84	≤0.001
1	0.607	0.678	150 / 85	≤0.001
2	0.610	0.689	152 / 91	≤0.001
3	0.605	0.682	159 / 91	≤0.001
4	0.598	0.693	152 / 85	≤0.001
mean	0.605 ± .005	0.683 ± .009	152 / 87	—
beats baseline	5/5	5/5	5/5	—

Table 3. Transformer across five training seeds, on the fixed article-disjoint split, versus the baseline (AP 0.527, matched-budget reference recall 0.606). "Beats baseline" counts seeds where the transformer wins that measure.

The paired transitions also refute a specific Version 1 claim — that the transformer leaves "the same subtle cases" undetected. Among the 264 truly propagandistic paragraphs, both models catch 110, the baseline alone catches 50, the transformer alone catches 34, and 70 are missed by both: the two models recover different cases, not the same ones. Among the 598 non-propagandistic paragraphs,

the transformer corrects 114 of the baseline's false positives while introducing 34 new ones, which is the source of its precision gain. Version 1's unsupported sentence about reclassifying "some fifty" particular cases is withdrawn in favour of these measured counts.

4. Interpretation

The reversal from Version 1 is instructive and worth stating plainly. The original claim — context improves precision but not recall — was an artifact of two things the corrected analysis removes: a leaky split, and a comparison made at each model's own default threshold rather than at equal cost. On the clean split, at equal review cost, context improves recall too; and the ranking and paired-correctness evidence, which do not depend on any threshold, both favour the transformer. The honest summary is therefore not "precision, not recall," but "a better detector overall, whose advantage is hidden if one compares only default operating points." What remains true from Version 1 is modesty about magnitude: the F1 gain at a fixed threshold is small and not significant, and the whole comparison rests on one training run per model.

5. Limitations

Several limitations bound these claims, and they matter. The effect is real but modest: the F1 difference at default thresholds is not significant, and the average-precision interval only just excludes zero. The transformer was trained across five seeds and the baseline is deterministic, so the central comparison no longer rests on one run; but five seeds share a single article-disjoint split, and split variation can exceed seed variation — the baseline alone, re-evaluated across ten different article-disjoint splits, ranges in propagandistic-class F1 from 0.43 to 0.52 (0.488 ± 0.025), which is wider than the seed spread, so the magnitudes here would be firmed up by repeating across splits, not only seeds. The model is also deliberately modest: DistilBERT at 256 tokens is not a strong 2026 model, and the comparison includes no larger transformer (RoBERTa, DeBERTa) and no hyperparameter ablations, so the size of the lexical-to-contextual gap under a stronger model is unknown. The matched-budget and ranking advantages are supported by intervals that exclude zero, but on one test set. As in Working Paper I, this is a classification result, not a validated measure of a construct: separating corpus-labelled propaganda from non-propaganda is not the same as measuring "manipulation density," and the score may partly track polarization, legitimate criticism or emotional tone. And a detector trained on human "propagandistic" labels must not be used to adjudicate which political positions are legitimate; it flags verifiable discursive features at most. The corpus is one journalistic source under one guideline set, so transfer to other languages, periods or political contexts is undemonstrated.

6. Conclusion

On a leakage-free, article-disjoint split, a fine-tuned DistilBERT is a better propaganda detector than the lexical baseline when the two are compared fairly — by ranking, at equal review cost, and by paired correctness — even though their single-operating-point F1 scores are not distinguishable. This corrects and reverses the earlier version's headline. It does not, however, validate the Politomorphism framework, establish that fine structural manipulation is now recovered, or show that "context helps" beyond reproducing one corpus's labels on one clean split. Those remain open. The strongest next steps are cross-split as well as cross-seed evaluation, a comparison against stronger contextual models with ablations, and — for the construct question that matters most — a purpose-built benchmark of contrastive pairs with independent annotation.

Reproducibility

The clean split, the baseline, the transformer run, and the comparison (bootstrap intervals, McNemar, matched-budget recall, paired transitions) are produced by the scripts in the clean_rerun/ directory of the repository, from the public PropXplain English partitions with fixed seeds. Per-paragraph scores for both models are saved there. Code, pinned dependencies, and both corrected papers are at github.com/politomorphism/politomorphism-defense. This version supersedes Version 1.

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